57 SITAR Portal Geoservices: Strategies for Sharing Strategic Data for Land Planning and Landscape Protection between Open Data and FLOSS Approaches

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SITAR (Special Superintendence ABAP of Rome) adopts a fully FLOSS-driven strategy for archaeological data interoperability and landscape protection. The Open Data portal relies on REST APIs and OGC standards to guarantee public access and reusability (CC BY-SA 4.0).

The backend infrastructure, hosted on the GARR network, relies on Debian Linux servers (including a backup instance). Data management and custom geospatial functions are handled via a PostgreSQL/PostGIS RDBMS instance. Data editing and structured input are carried out through QGIS desktop environments, supporting versions from 3.40-3.XX up to 4.0-4.XX.

Massive datasets — from core SITAR layers to historical raster/vector maps (1924 Agro Romano map, PTRS for instance) — are published via the open-source map server GeoServer, which delivers layers, styles, and metadata. The WordPress frontend integrates a custom module that directly parses GetCapabilities requests from GeoServer, dynamically populating the Geoservices page with metadata and connection endpoints. All created datasets are publicly accessible and queryable directly within the SITAR WebGIS platform. Moreover, they are available to any OGC-compliant desktop or web GIS software: OGC WMS allows seamless visualization and consultation, while OGC WFS enables direct vector feature download (GeoJSON, Shapefile, CSV, KML).

The paper presents case studies on combining these open datasets, demonstrating the strategic value of an end-to-end FLOSS architecture for heritage protection, the reconstruction of archaeological depositional dynamics, and advanced landscape analysis — a qualitative and quantitative potential on which the working group has focused in recent years.